PROBLEM VF20

Cylinder Levitated by a Jet

Original statement · VF20.pdf

Radial Poiseuille flow | Thin film | Force balance

OFFICIAL PROBLEM STATEMENT

FLUID MECHANICS I. PROBLEM 2 (10 POINTS). 16-01-2019. Time: 75 minutes. The cylinder of radius R and height $2R$ shown in the figure is submerged in an infinite tank of liquid of density ρ and viscosity μ. The cylinder, of density $\rho_s > \rho$, is sustained at a constant height $h \ll R$ by the effect of a narrow jet of volumetric flow rate Q and radius of the order of h, injected at the center of the base of the cylinder (see the sketch). Assuming the fluid movement is negligible in the bulk of the tank (P_0 = const.) and the flow below the cylinder is dominated by viscosity, we want to determine the dependence of h as a function of Q. To this end, the following sequence is suggested:

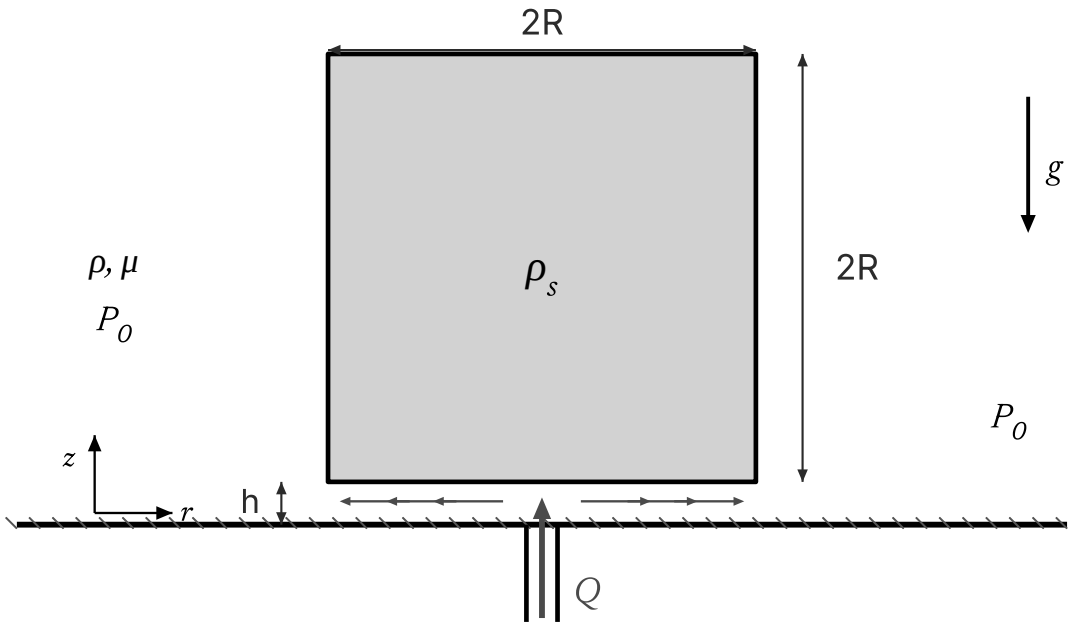
1. Give the criterion, as a function of the known parameters, for the flow in the gap of width h between the cylinder and the bottom of the tank to be dominated by viscosity.

2. Determine the velocity profile as a function of the local reduced pressure gradient $\partial P/\partial r$.

3. Obtain the local flow rate $q(r)$ at a generic radial position r

4. Use the continuity equation to derive an equation for the reduced pressure P, with all needed boundary conditions. Obtain P.

5. Determine h



Statement figure (redrawn from VF20.pdf): a cylinder of radius R, height 2R and density ρ_s hovering at the height h above the bottom of a tank of liquid (ρ, μ, pressure P_0 far away) on a jet of flow rate Q injected at the centre of its base; z is vertical and r radial

Source: VF20.pdf (page 1: statement; pages 2 and 3: handwritten key; exam of 16-01-2019). Notes.pdf, Chapter 7: slender films, Eqs. 7.67-7.71; the wedge effect, Eq. 7.72.

Worked solution

Hypotheses, Data and Unknowns

- The gap between the base of the cylinder ($z = h$) and the bottom of the tank ($z = 0$) is slender, axisymmetric and steady; the liquid enters it through the jet at the centre ($r \lesssim h$) and leaves at $r = R$ to the tank, where the reduced pressure is P_0 (the liquid is at rest there).
- Reduced pressure $P = p + \rho g z$. The cylinder is in equilibrium: its weight $W = 2\pi R^3 \rho_s g$ (volume $2\pi R^3$) balances the force of the liquid.

Symbol	Meaning	SI unit
R, $2R$	radius and height of the cylinder	m
ρ_s, ρ, μ	density of the cylinder, of the liquid, viscosity	kg/m^3, kg/m^3, kg/(m s)

Symbol	Meaning	SI unit
h, Q	height of the gap, flux of the jet	m, m^3/s

Criterion for Viscous Dominance

The convective acceleration is negligible against the viscous force (Eq. 7.41) when $\frac{\rho V_{zc} h}{\mu}\frac{h}{R} \ll 1$, with V_{zc} the velocity scale of the jet, of radius $\sim h$, so $V_{zc} \sim Q/h^2$. The radial velocity in the gap follows from the flux, $v_r = Q/(2\pi r h)$, which is $\sim Q/h^2 \sim V_{zc}$ at the entrance $r \sim h$. Then

$$\frac{\rho V_{zc} h}{\mu}\,\frac{h}{R} \sim \frac{\rho Q}{\mu h}\,\frac{h}{R} = \frac{\rho Q}{\mu R} \ll 1$$

(the key states $V_r \sim Q/h^2$ for $r \sim h$ and boxes the result). This is the estimate of the key, with the global slenderness h/R. More locally, at a radius r the ratio of convective to viscous terms is $\rho v_r h^2/(\mu r) = \rho Q h/(2\pi\mu r^2)$, which decreases with r: it is largest near the entrance, where it is of order $\rho Q/(\mu h)$ (stricter than the boxed $\rho Q/(\mu R)$ by R/h), and the slender-film solution below is valid for $r \gg h$ when $\rho Q h/(\mu r^2) \ll 1$ there.

Velocity Profile and Flux

Radial momentum in the gap (Eq. 7.69 in cylindrical coordinates), with no slip at the bottom of the tank ($z = 0$) and at the base of the cylinder ($z = h$):

$$0 = -\frac{\partial P}{\partial r} + \mu\frac{\partial^2 v_r}{\partial z^2}; \qquad z = 0:\ v_r = 0, \qquad z = h:\ v_r = 0$$

Integrating twice, $v_r = \frac{1}{2\mu}\frac{\partial P}{\partial r} z^2 + Az$ ($B = 0$ by the condition at $z = 0$), and $0 = \frac{1}{2\mu}\frac{\partial P}{\partial r} h^2 + Ah$ gives A:

$$v_r = -\frac{1}{2\mu}\,\frac{\partial P}{\partial r}\left(hz - z^2\right)$$

The local flux through the cylinder of radius r and height h is

$$q(r) = \int_0^h 2\pi r\, v_r\, dz = -\frac{2\pi r}{2\mu}\frac{\partial P}{\partial r}\left[\frac{hz^2}{2} - \frac{z^3}{3}\right]_0^h = -\frac{\pi h^3 r}{6\mu}\frac{\partial P}{\partial r}$$

Pressure

Continuity in the gap: no liquid enters or leaves through the bottom or the base (both are walls), so $q(r) = Q$ for every r:

$$Q = -\frac{\pi h^3 r}{6\mu}\frac{\partial P}{\partial r} \implies \frac{\partial P}{\partial r} = -\frac{6\mu Q}{\pi h^3 r}$$

Integrating in r with the boundary condition $P = P_0$ at $r = R$:

$$P = P_0 - \frac{6\mu Q}{\pi h^3}\ln\frac{r}{R}$$

an overpressure for $r < R$, which grows (logarithmically) towards the centre, where the slender approximation stops being valid (at $r \sim h$).

Height of the Cylinder

The force of the liquid on the cylinder is the difference of the pressures on its base and on its top, over the area πR^2 (the lateral pressure is horizontal). At the top, $z = h + 2R$, the liquid is at rest with the reduced pressure P_0, so $p_t = P_0 - \rho g(h + 2R)$; at the base, $p_b = P - \rho g h$:

$$F = \int_0^R 2\pi r\,(p_b - p_t)\,dr = \int_0^R 2\pi r\,[2\rho g R + (P - P_0)]\,dr = 2\pi\rho g R^3 - \frac{12\mu Q}{h^3}\int_0^R r\ln\frac{r}{R}\,dr$$

With $r = Rs$, $\int_0^R r \ln \frac{r}{R} dr = R^2 \int_0^1 s \ln s \, ds = R^2 \left[\frac{s^2}{2} \ln s - \frac{s^2}{4} \right]_0^1 = -\frac{R^2}{4}$:

$$F = 2\pi \rho g R^3 + \frac{3\mu Q R^2}{h^3}$$

The first term is the buoyancy (the weight of the displaced liquid, $\rho g \cdot 2\pi R^3$) and the second is the lift of the jet. The cylinder floats at the height h when $F = W = 2\pi R^3 \rho_s g$:

$$\frac{3\mu Q R^2}{h^3} = 2\pi R^3 g \left(\rho_s - \rho \right) \implies h = \left[\frac{3\mu Q R^2}{2\pi R^3 g \left(\rho_s - \rho \right)} \right]^{1/3} = \left[\frac{3\mu Q}{2\pi R g \left(\rho_s - \rho \right)} \right]^{1/3}$$

Answers and Checks

- Viscous dominance: $\rho Q / (\mu R) \ll 1$ (and $h \ll R$).
- Profile: $v_r = -\frac{1}{2\mu} \frac{\partial P}{\partial r} (hz - z^2)$; flux: $q = -\frac{\pi h^3 r}{6\mu} \frac{\partial P}{\partial r} = Q$; pressure: $P = P_0 - \frac{6\mu Q}{\pi h^3} \ln \frac{r}{R}$.
- Height: $h = \left[\frac{3\mu Q}{2\pi R g (\rho_s - \rho)} \right]^{1/3}$.
- Dimensions: $\mu Q / (R g \Delta \rho)$ is $ML^{-1}T^{-1} \cdot L^3 T^{-1} / (L \cdot LT^{-2} \cdot ML^{-3}) = L^3$, so h is a length; $\rho Q / (\mu R)$ is $ML^{-3} L^3 T^{-1} / (ML^{-1} T^{-1} L) = 1$.
- Limits: $Q \to 0$ gives $h \to 0$ (the cylinder rests on the bottom) and $h \propto Q^{1/3}$; $\rho_s \to \rho$ makes the cylinder neutrally buoyant and $h \to \infty$ (it floats with any Q).
- Check against the official key: VF20.pdf gives the same criterion (stated without the step $V_{zc} \sim Q / h^2$), profile, q, P, force ($2\pi \rho g R^3 + 3\mu Q R^2 / h^3$) and h (left as $\left[3\mu Q R^2 / (2\pi R^3 g (\rho_s - \rho)) \right]^{1/3}$).